



US 20250276771A1

(19) **United States**

(12) **Patent Application Publication**
Wieland

(10) **Pub. No.: US 2025/0276771 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **SYSTEMS AND APPROACHES FOR
ASSEMBLING A MARITIME VEHICLE**

(71) Applicant: **SARONIC TECHNOLOGIES**, Austin,
TX (US)

(72) Inventor: **Dana Wieland**, Austin, TX (US)

(21) Appl. No.: **19/067,320**

(22) Filed: **Feb. 28, 2025**

Related U.S. Application Data

(60) Provisional application No. 63/561,282, filed on Mar.
4, 2024.

Publication Classification

(51) **Int. Cl.**
B63B 73/20 (2020.01)
B63B 73/40 (2020.01)

B63B 73/43 (2020.01)

B63B 73/46 (2020.01)

B63B 73/50 (2020.01)

B63B 73/60 (2020.01)

(52) **U.S. Cl.**

CPC **B63B 73/20** (2020.01); **B63B 73/43**

(2020.01); **B63B 73/46** (2020.01); **B63B 73/48**

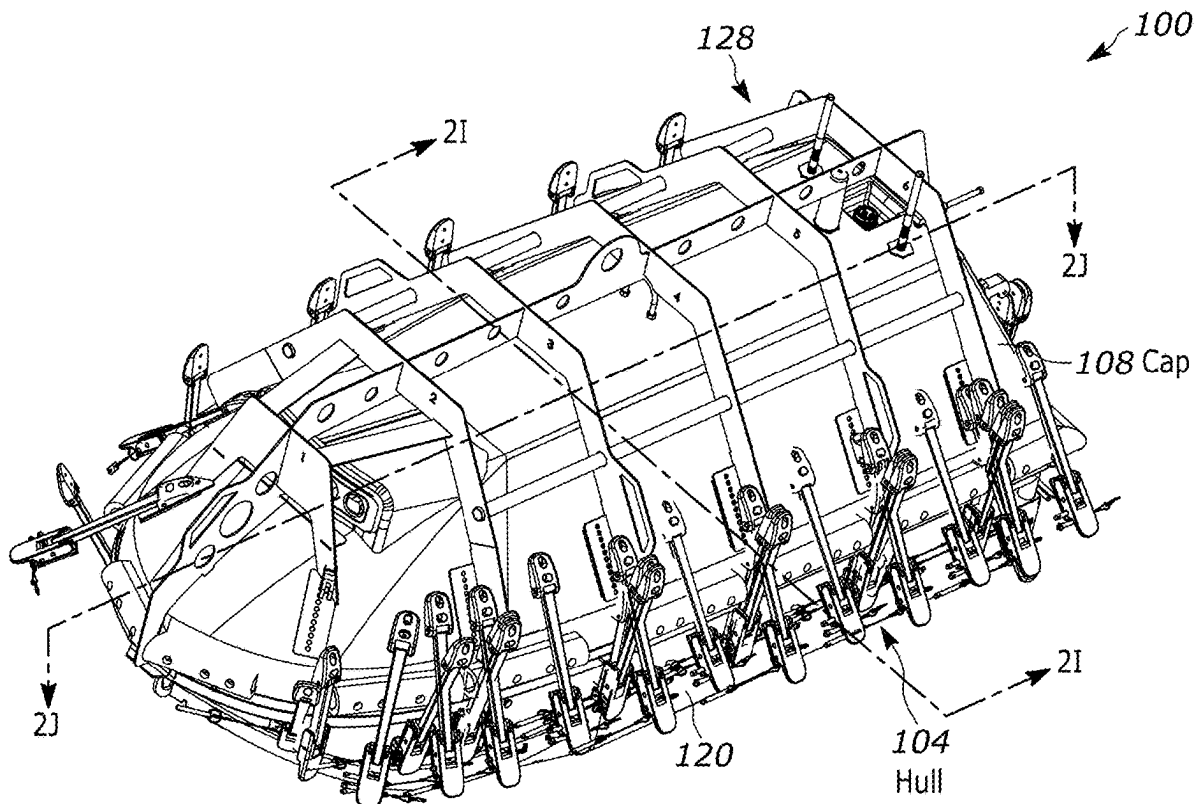
(2020.01); **B63B 73/50** (2020.01); **B63B 73/60**

(2020.01); **B63B 2231/10** (2013.01)

(57)

ABSTRACT

An approach for assembling a maritime vehicle includes forming a keel panel from a single sheet of metal, forming a first side panel from a single sheet of metal, forming a second side panel from a single sheet of metal, and forming a lip member. The first side panel and the second side panel are temporarily coupled with the lip member. The keel panel is secured with the first side panel and the second side panel using a cold joining process.



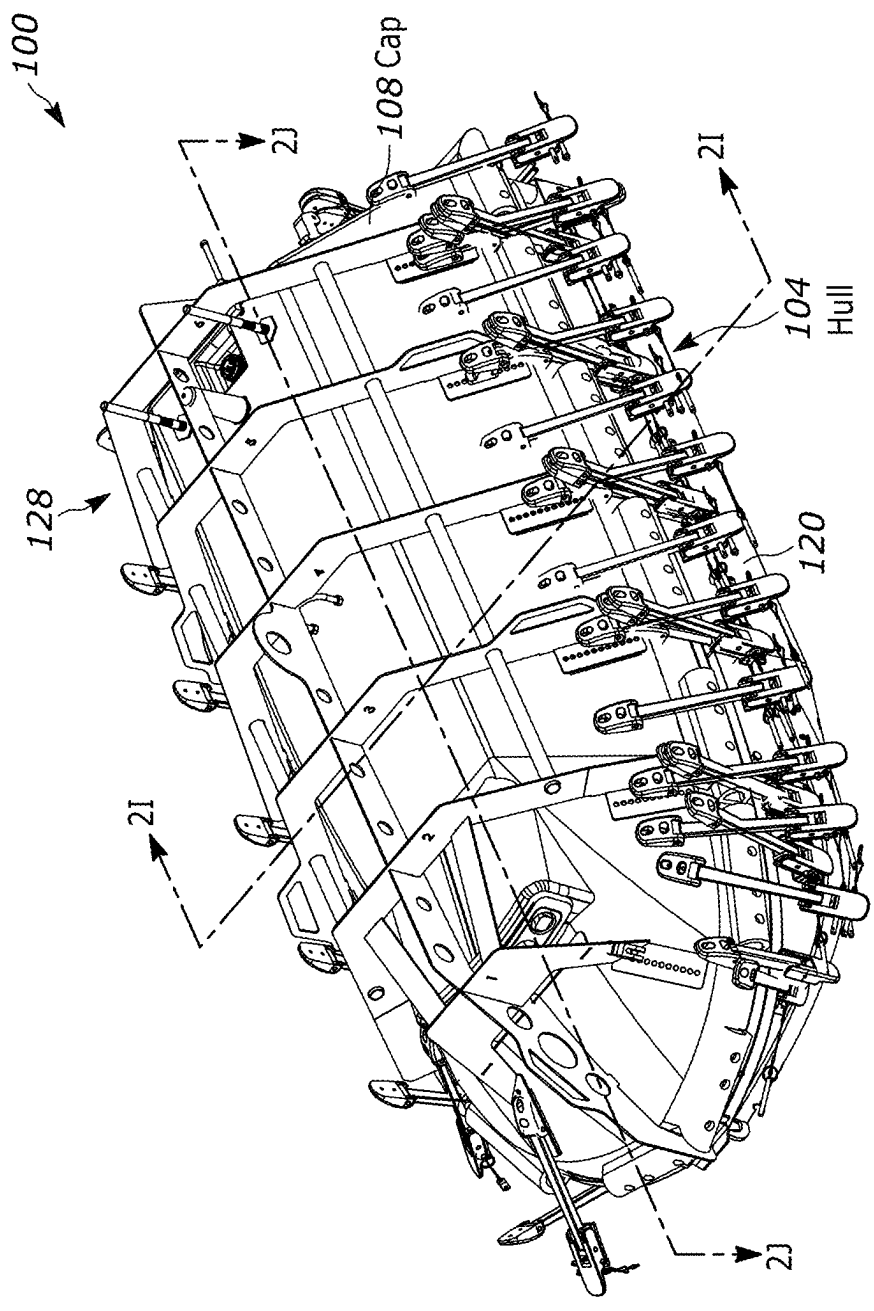


FIG. 1A

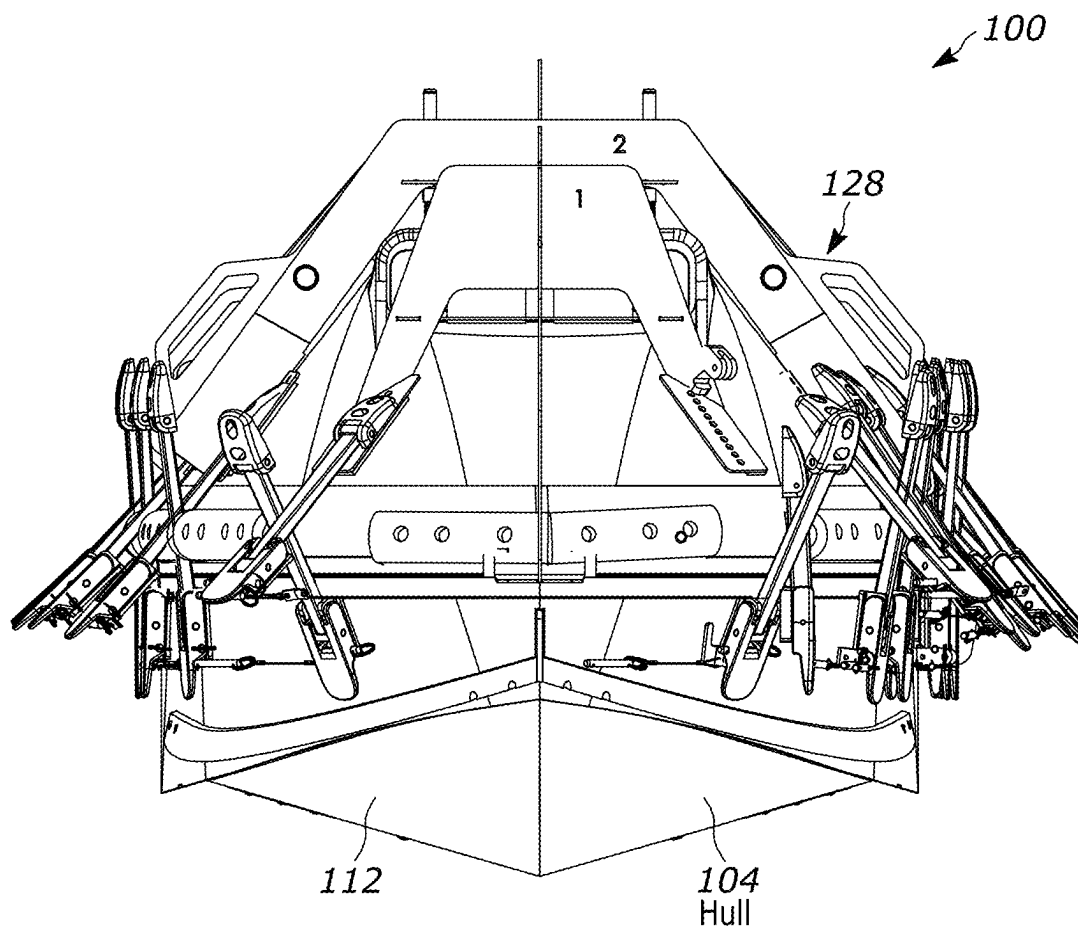


FIG. 1B

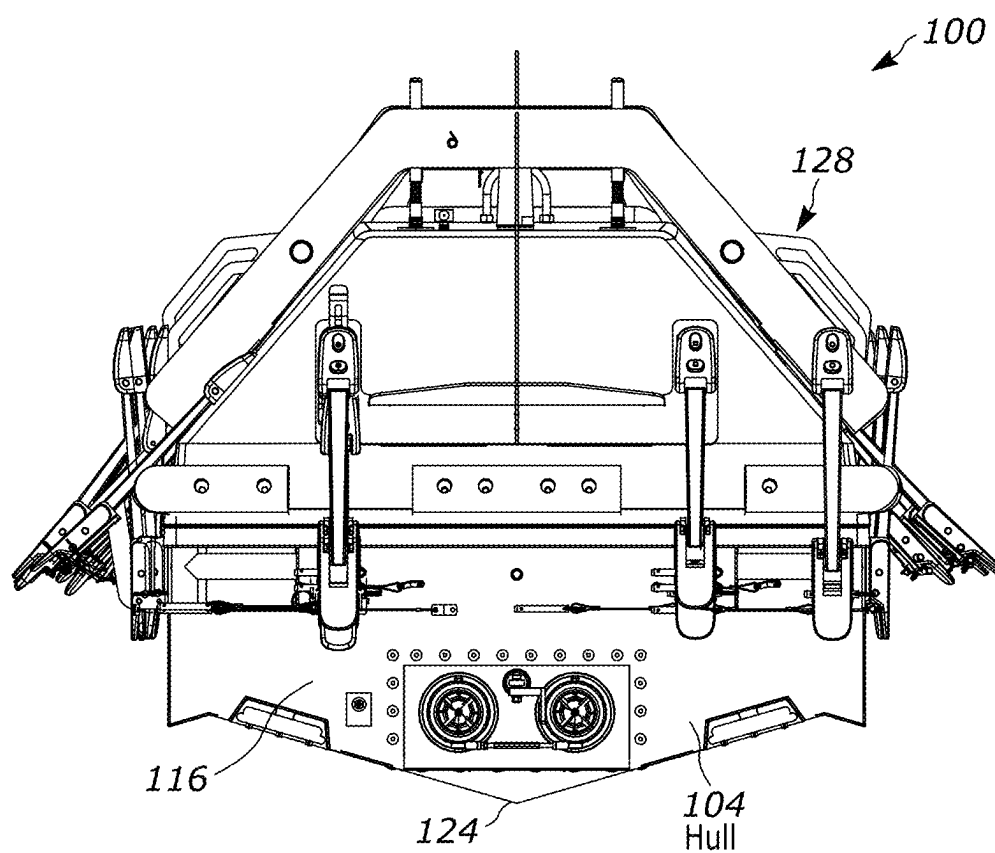


FIG. 1C

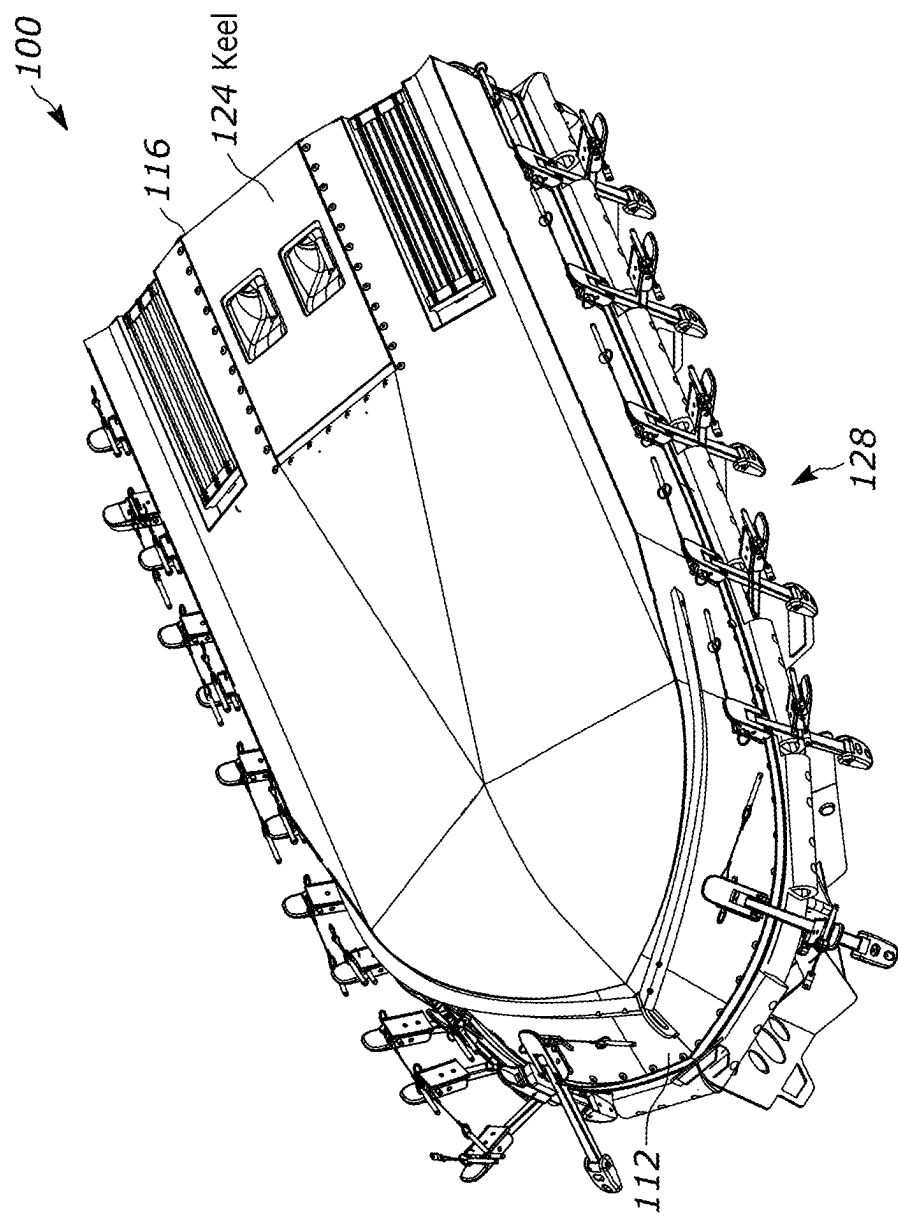


FIG. 1D

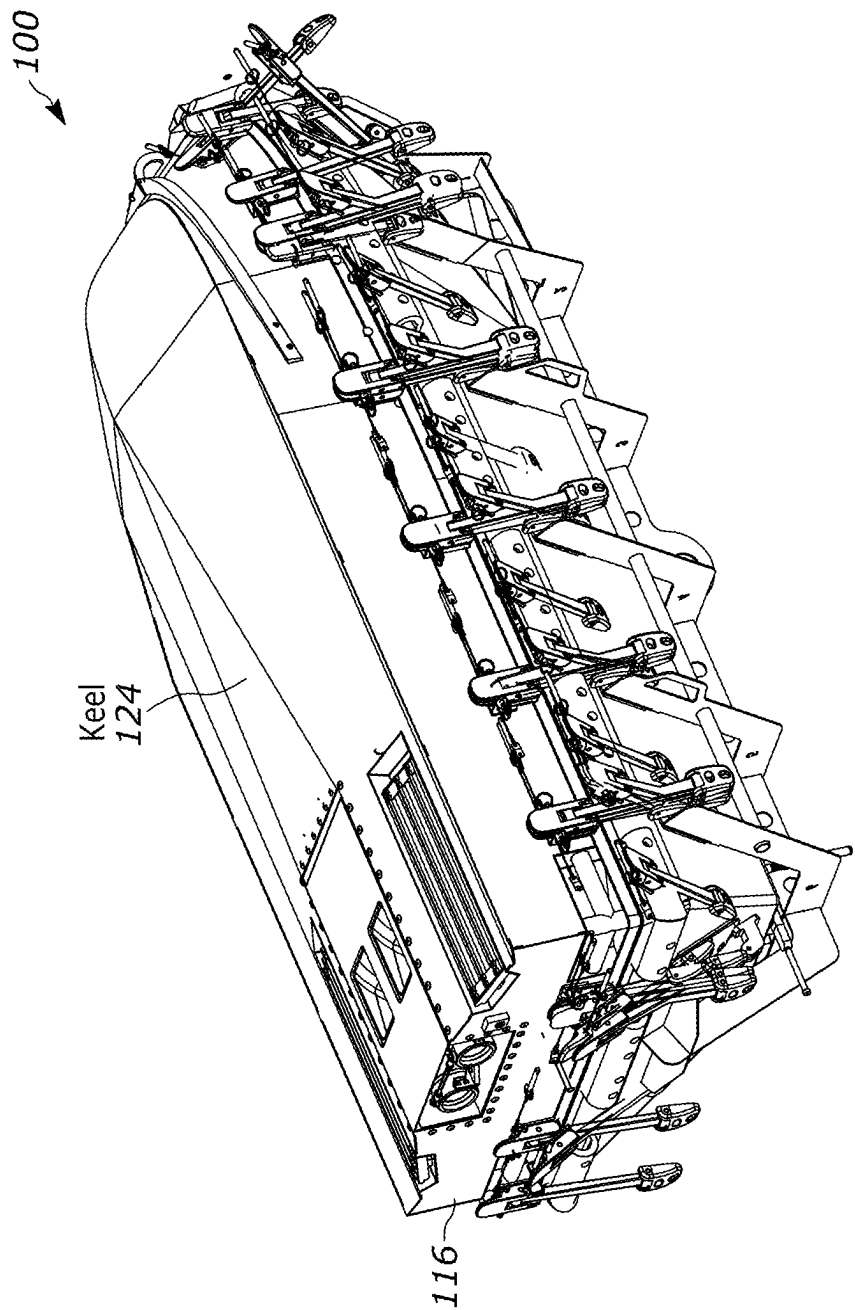


FIG. 1E

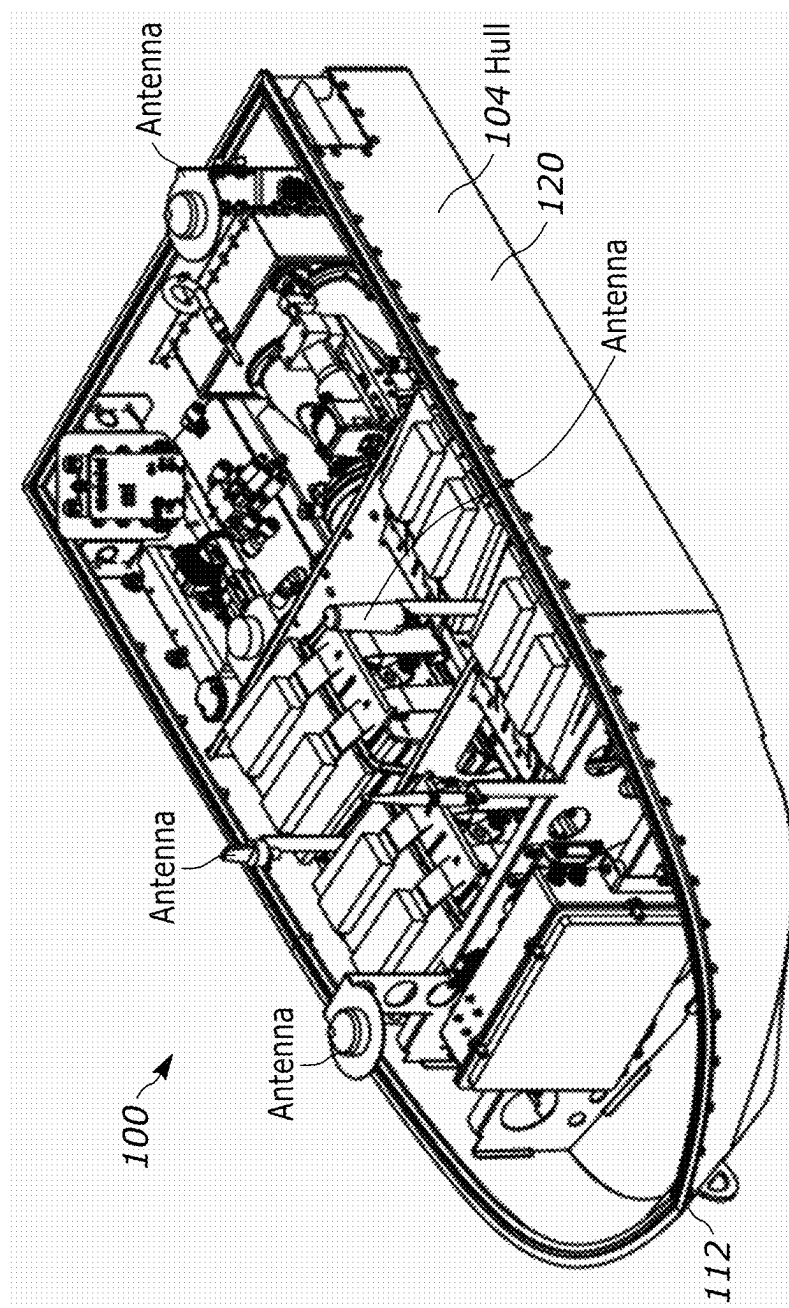


FIG. 1F

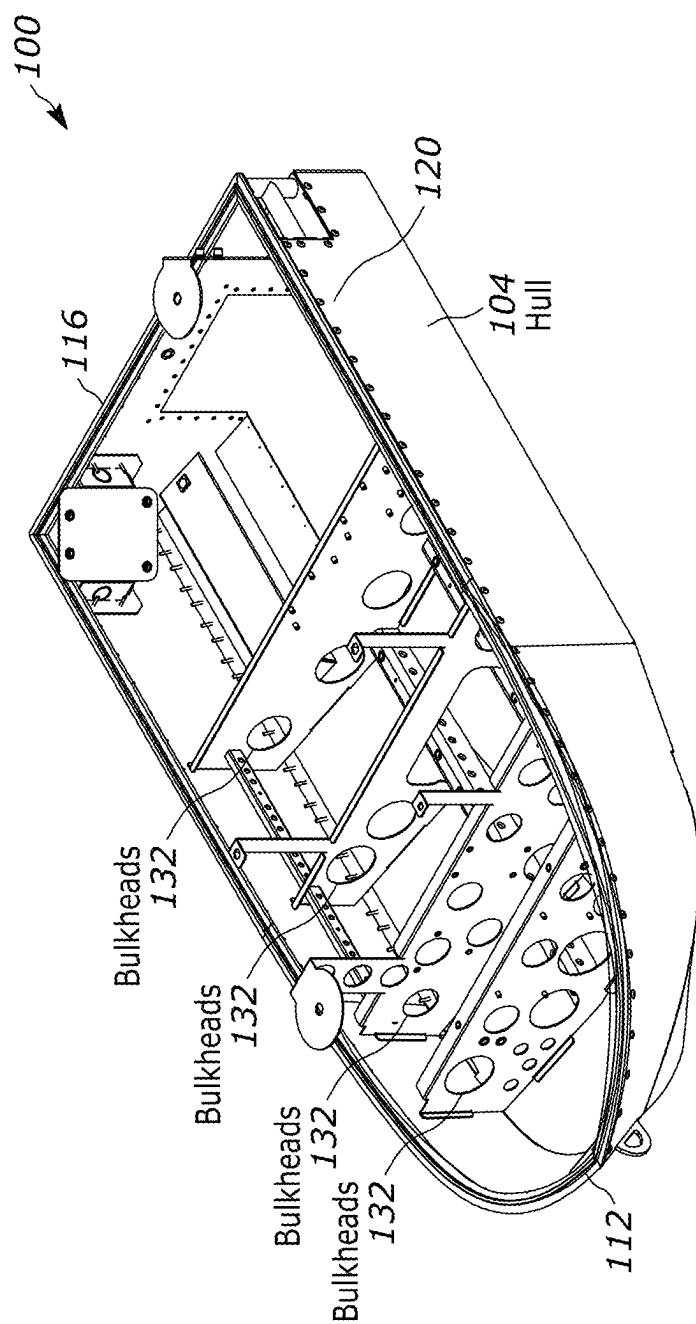


FIG. 1G

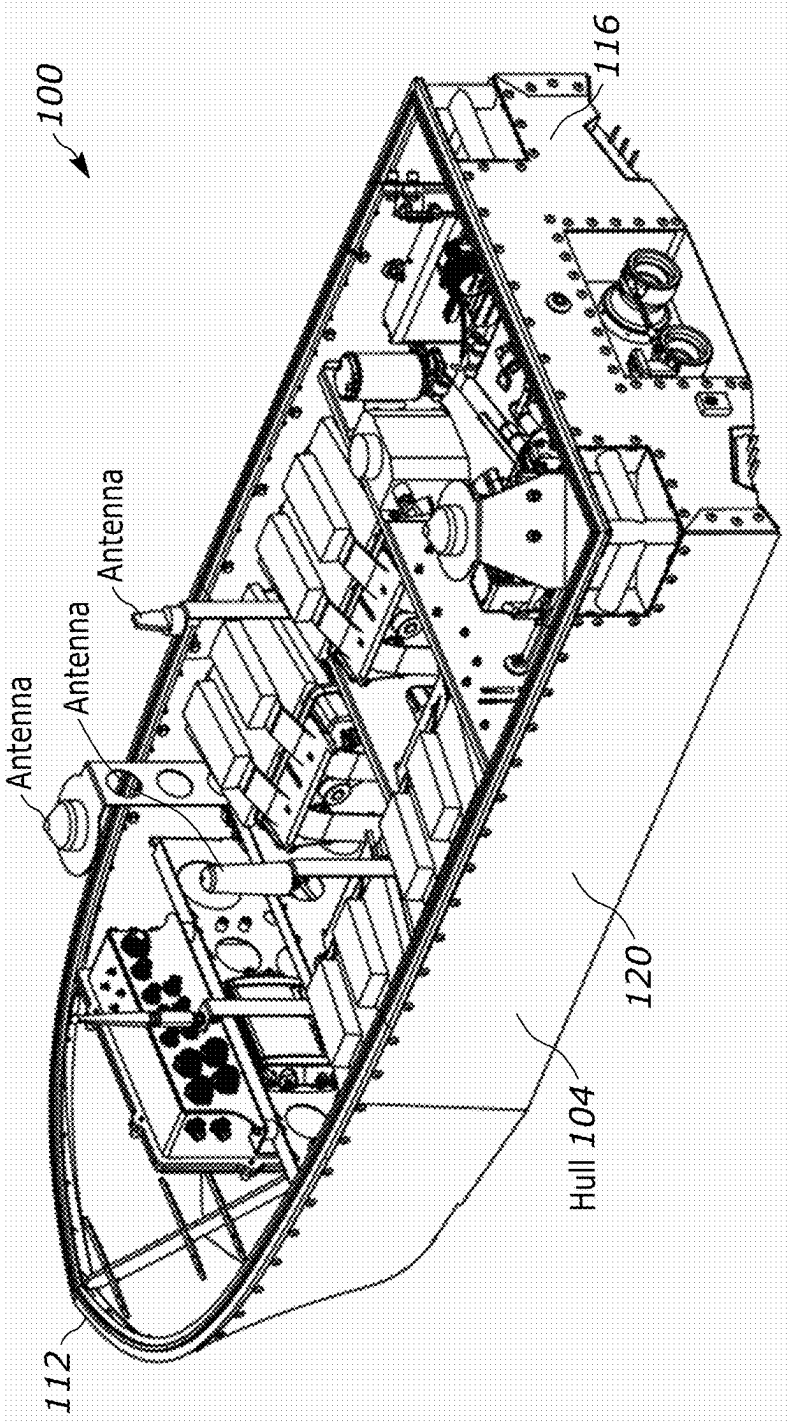


FIG. 1H

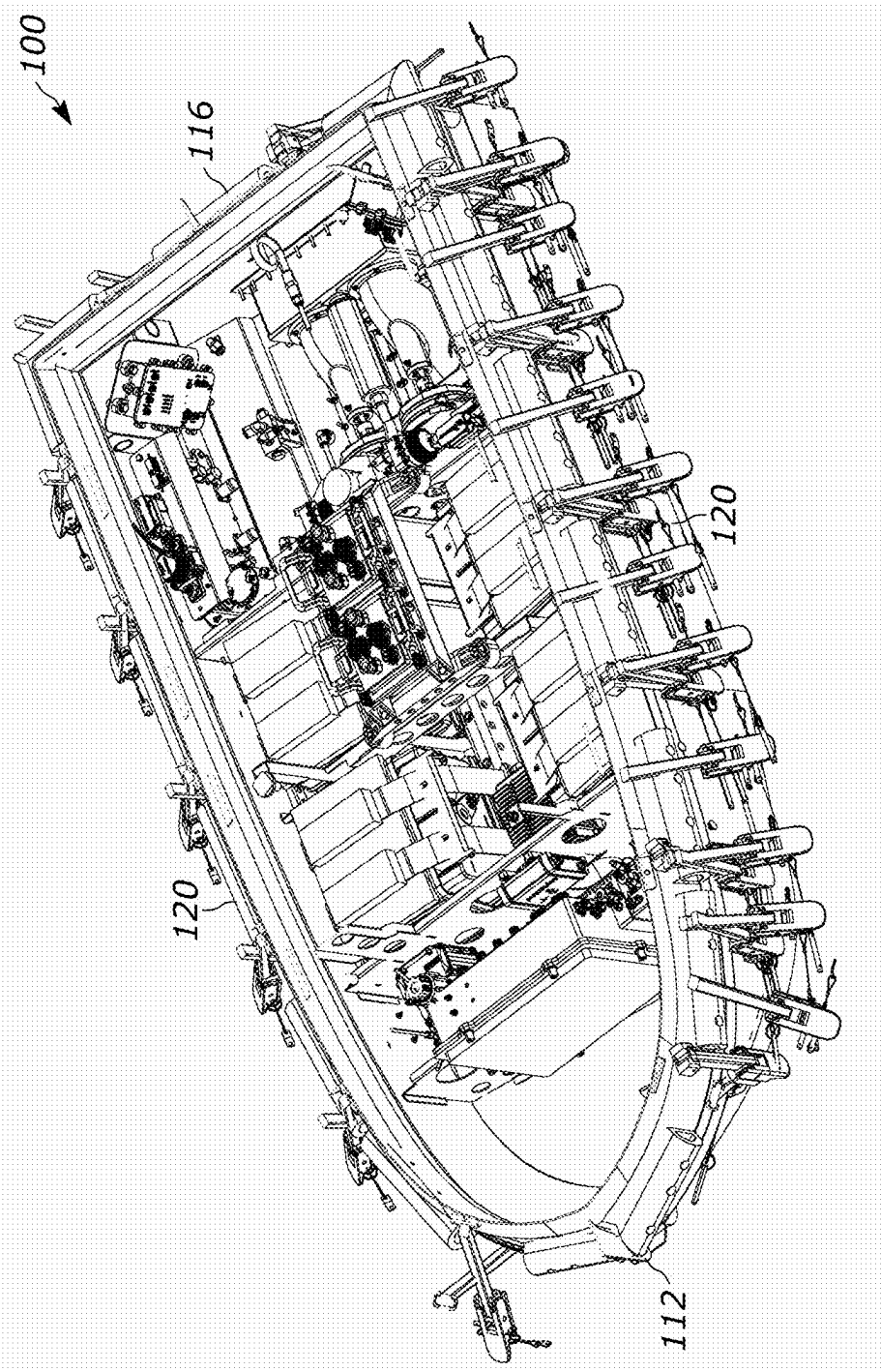


FIG. 11

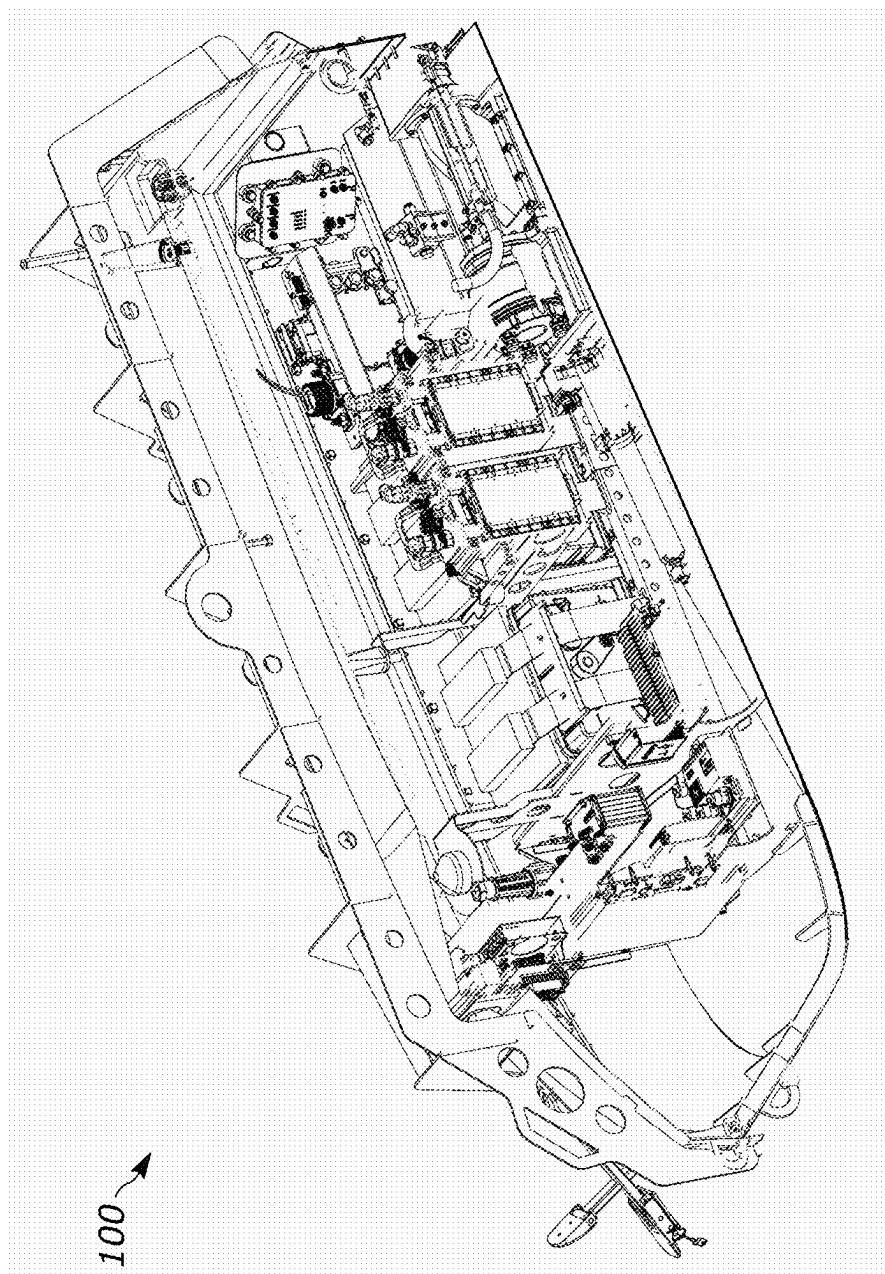


FIG. 1J

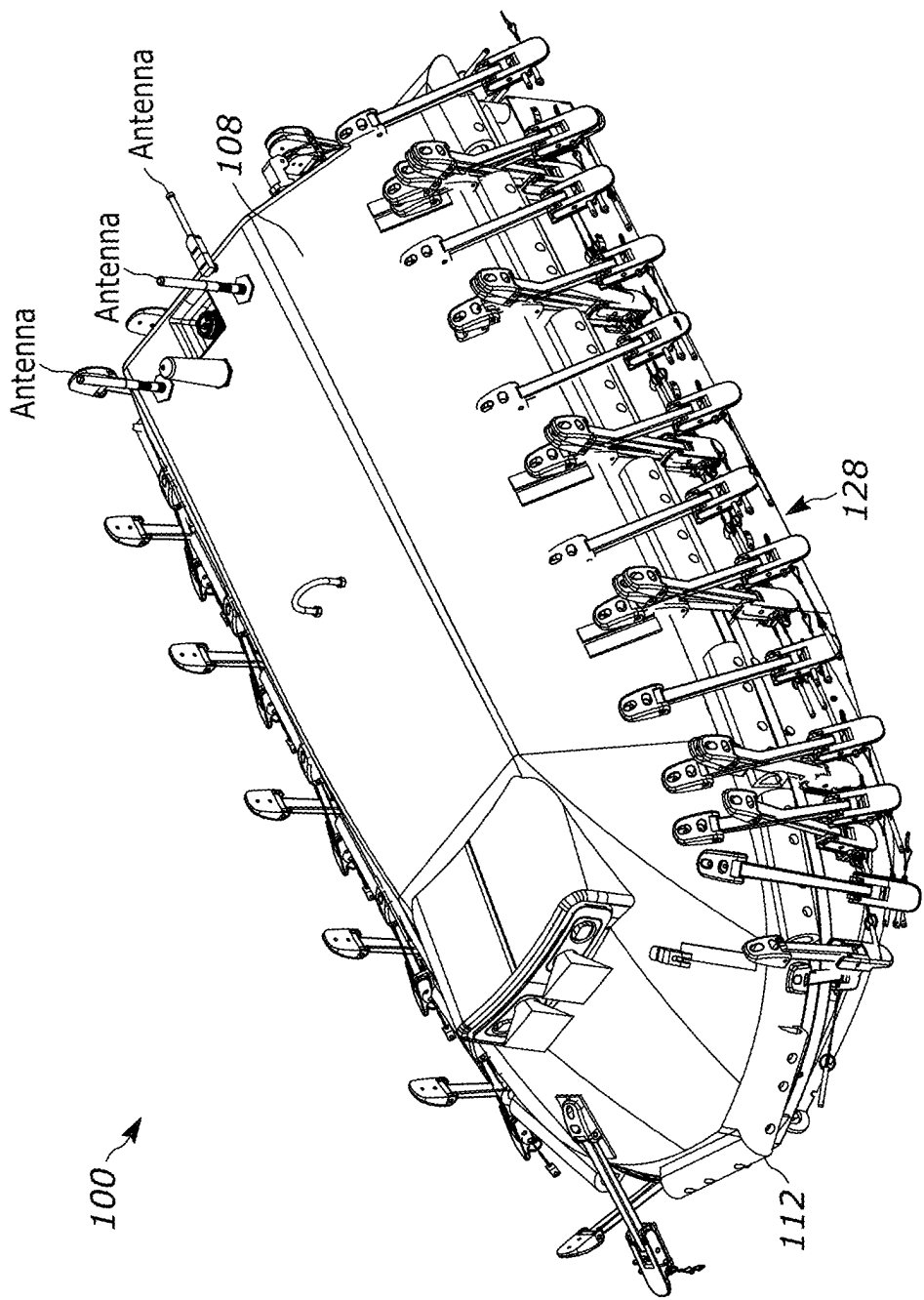


FIG. 1K

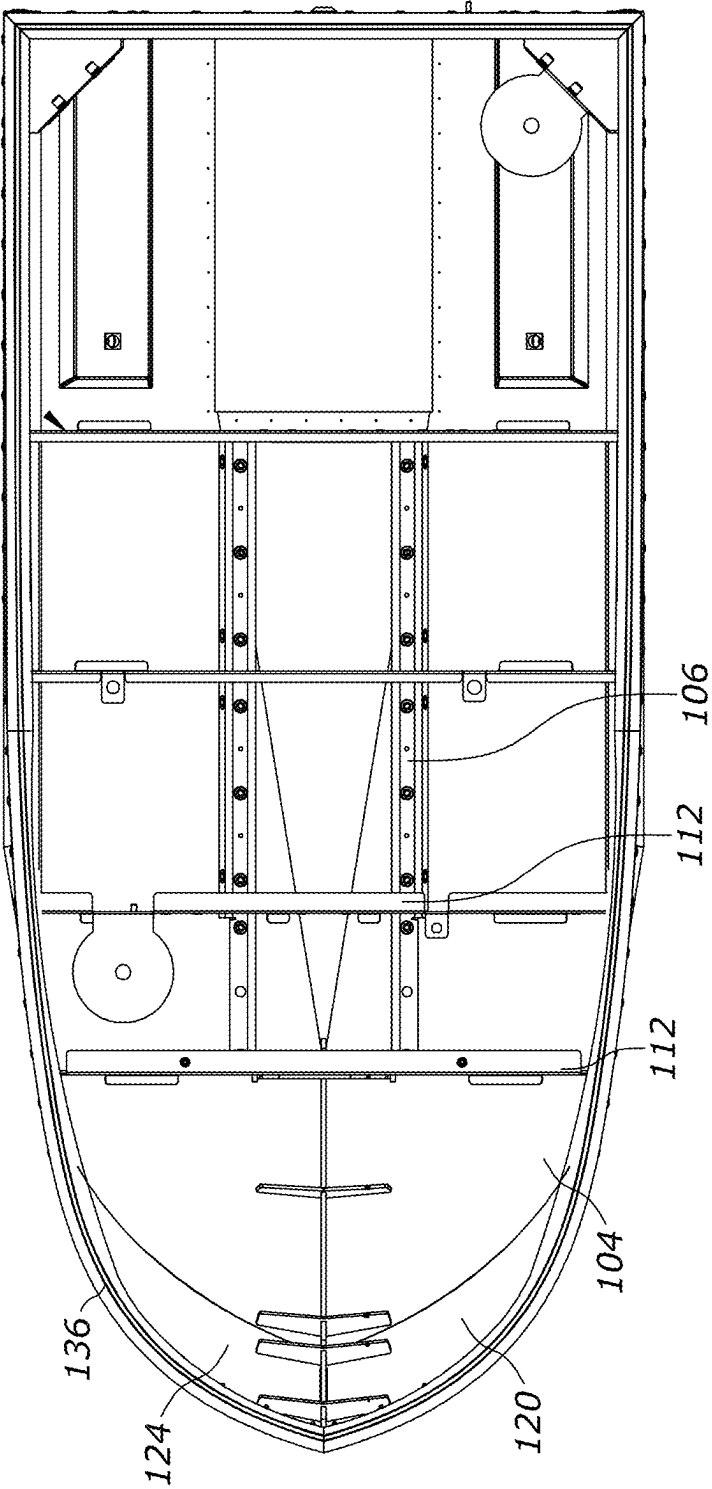


FIG. 2

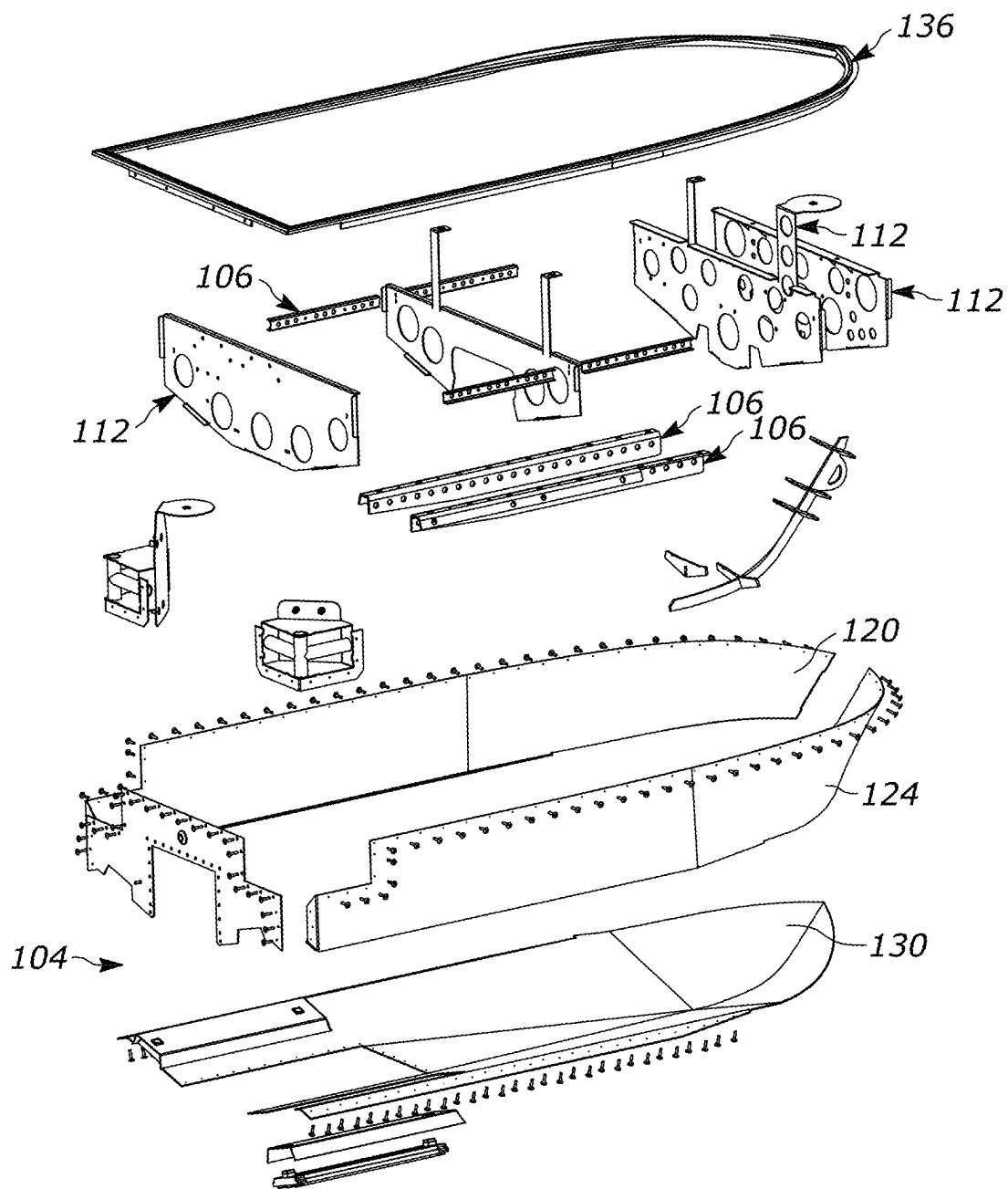


FIG. 3

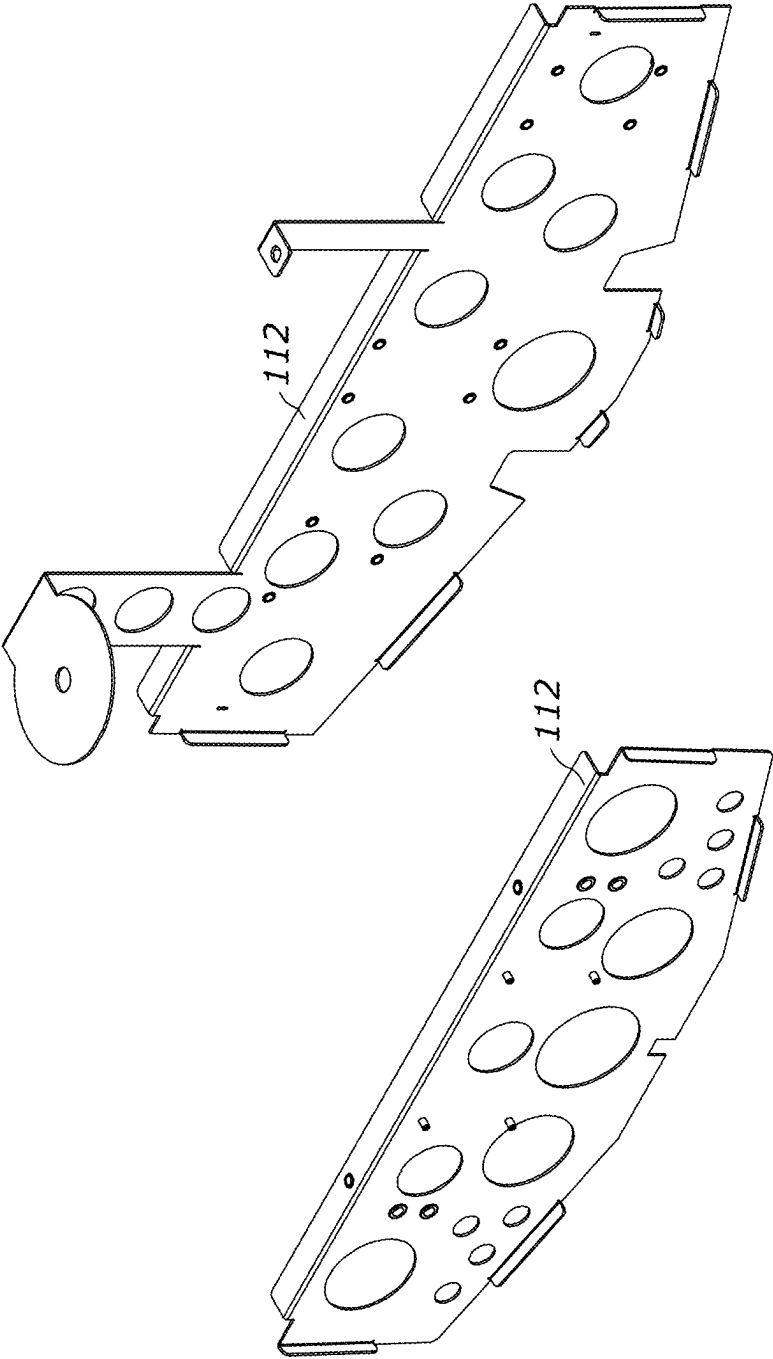


FIG. 4

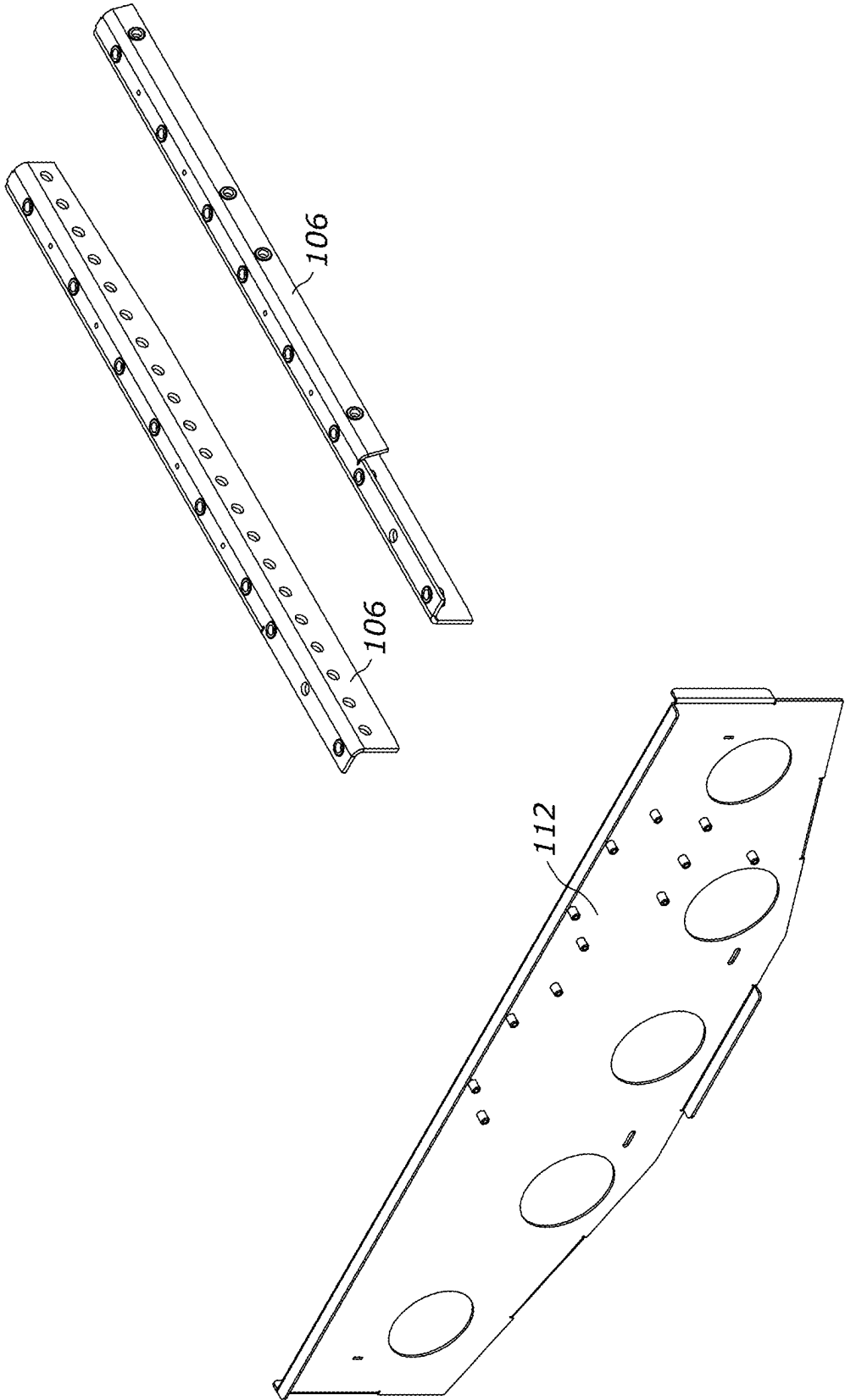


FIG. 5

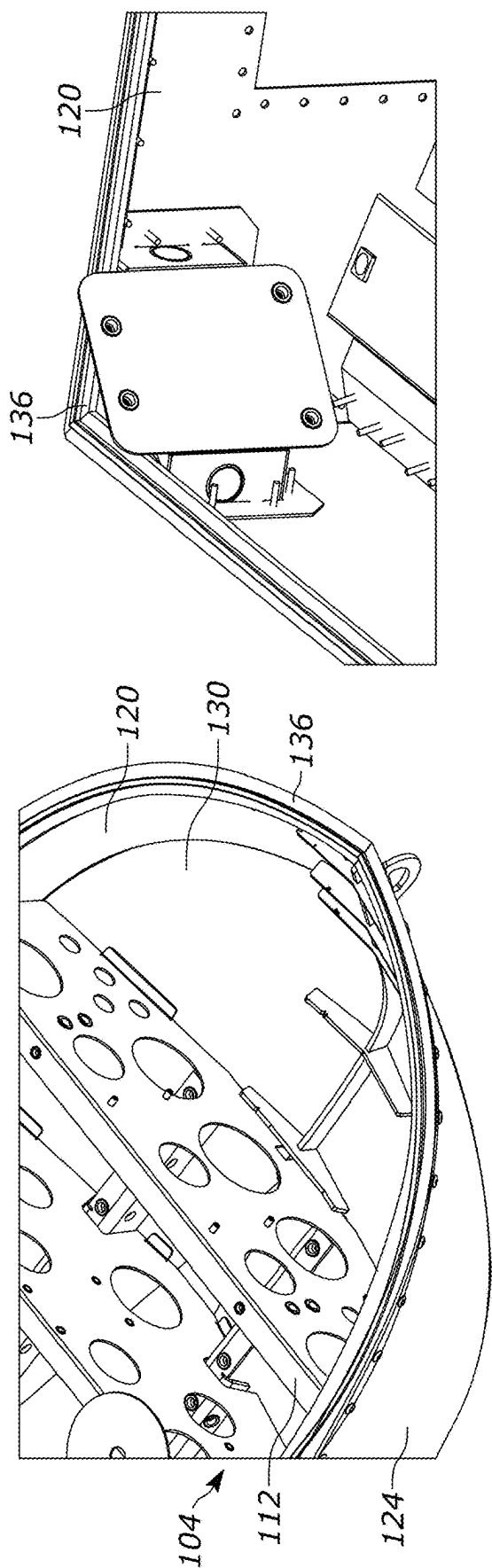


FIG. 6

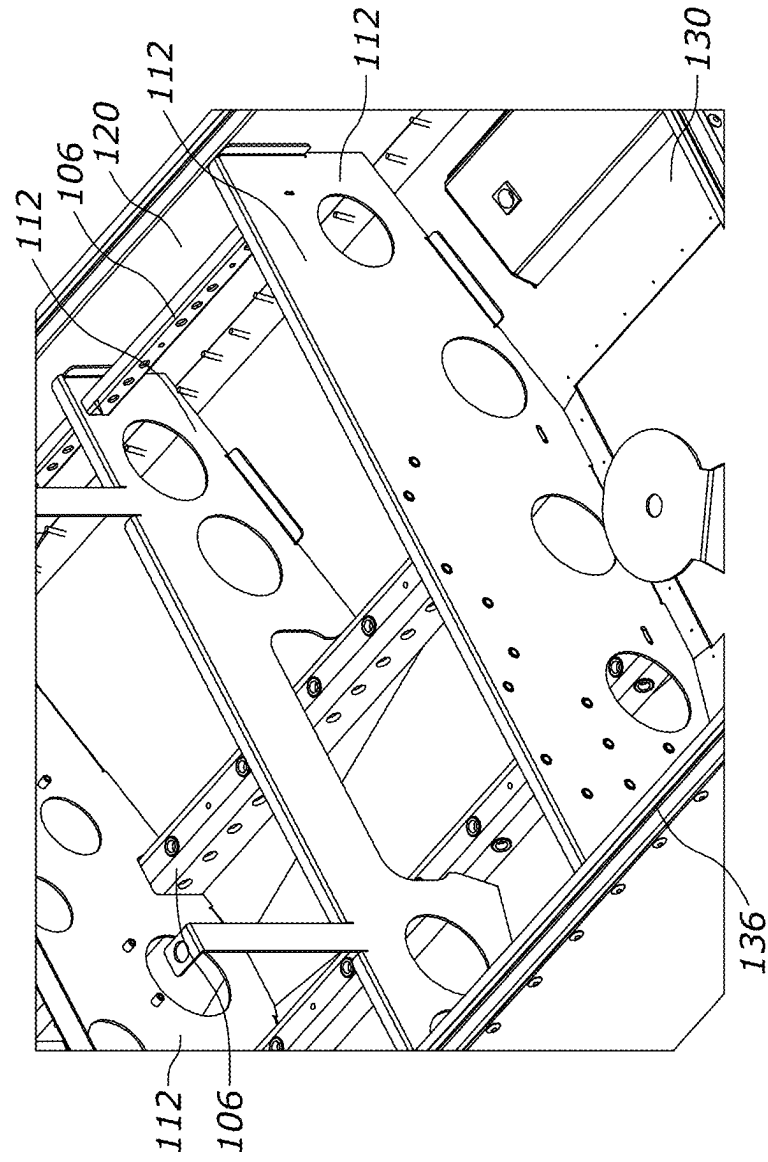


FIG. 7

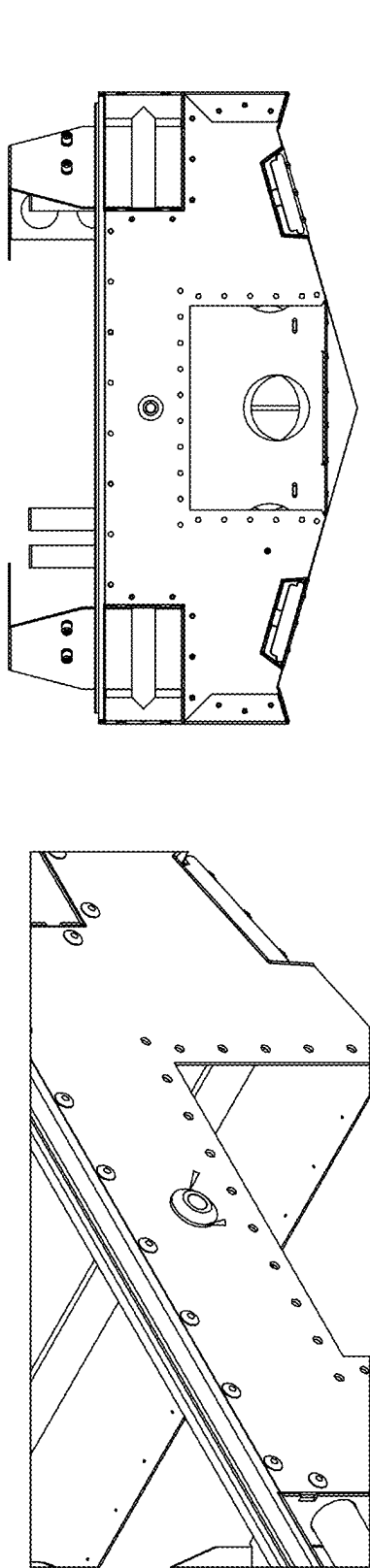


FIG. 8

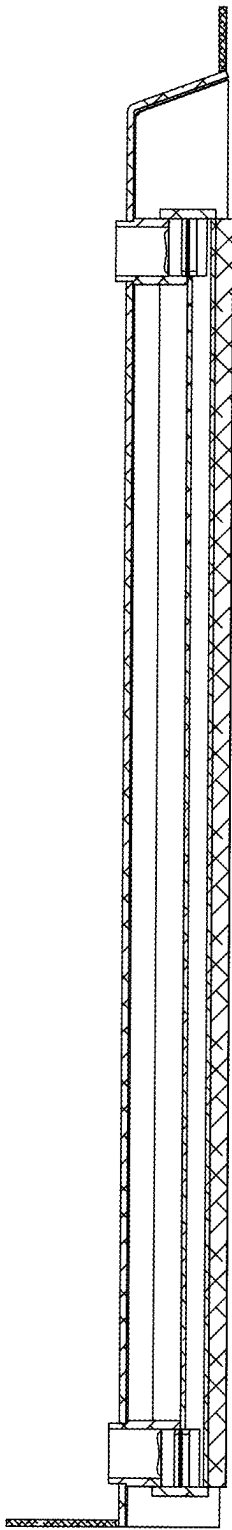


FIG. 9

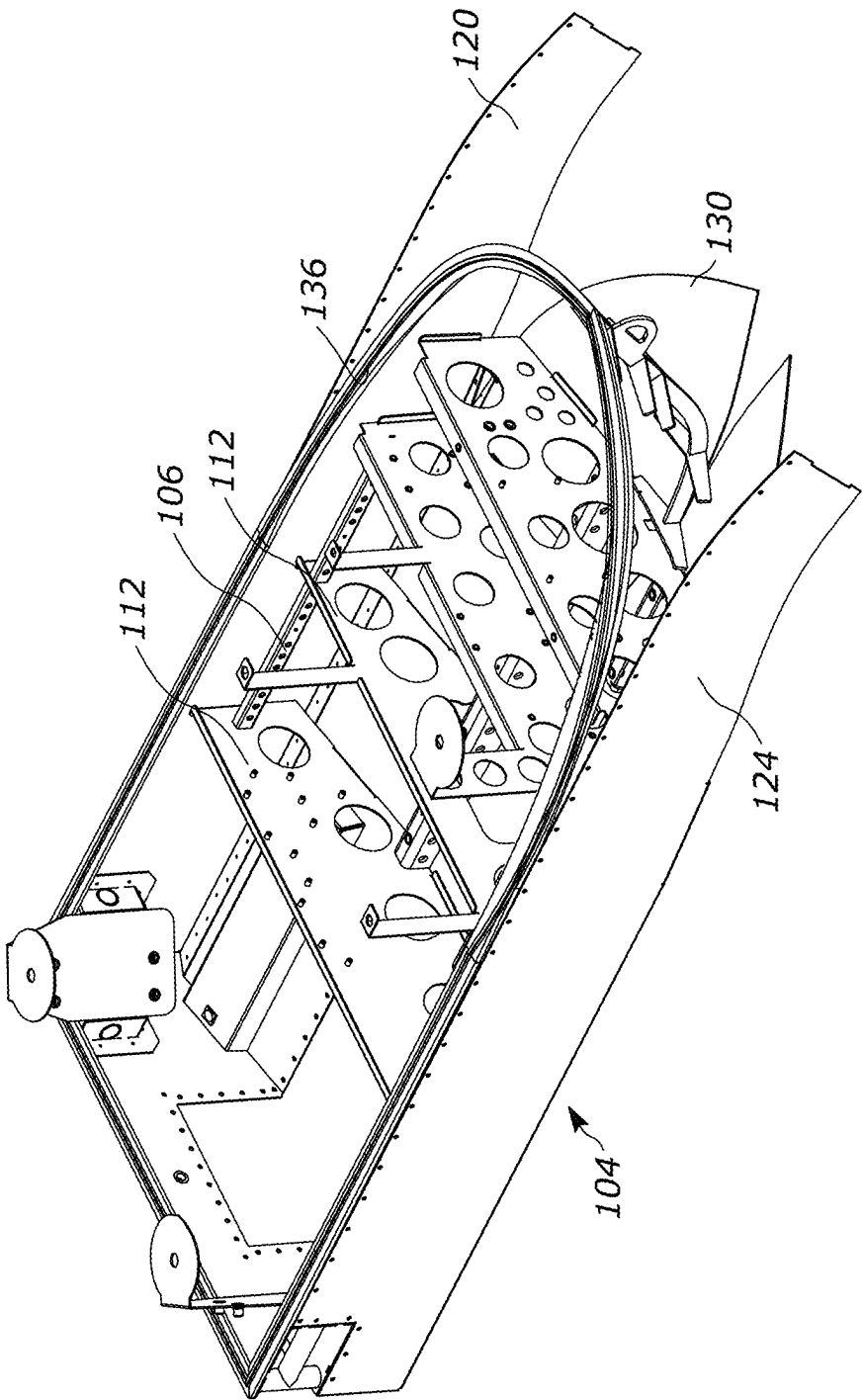


FIG. 10

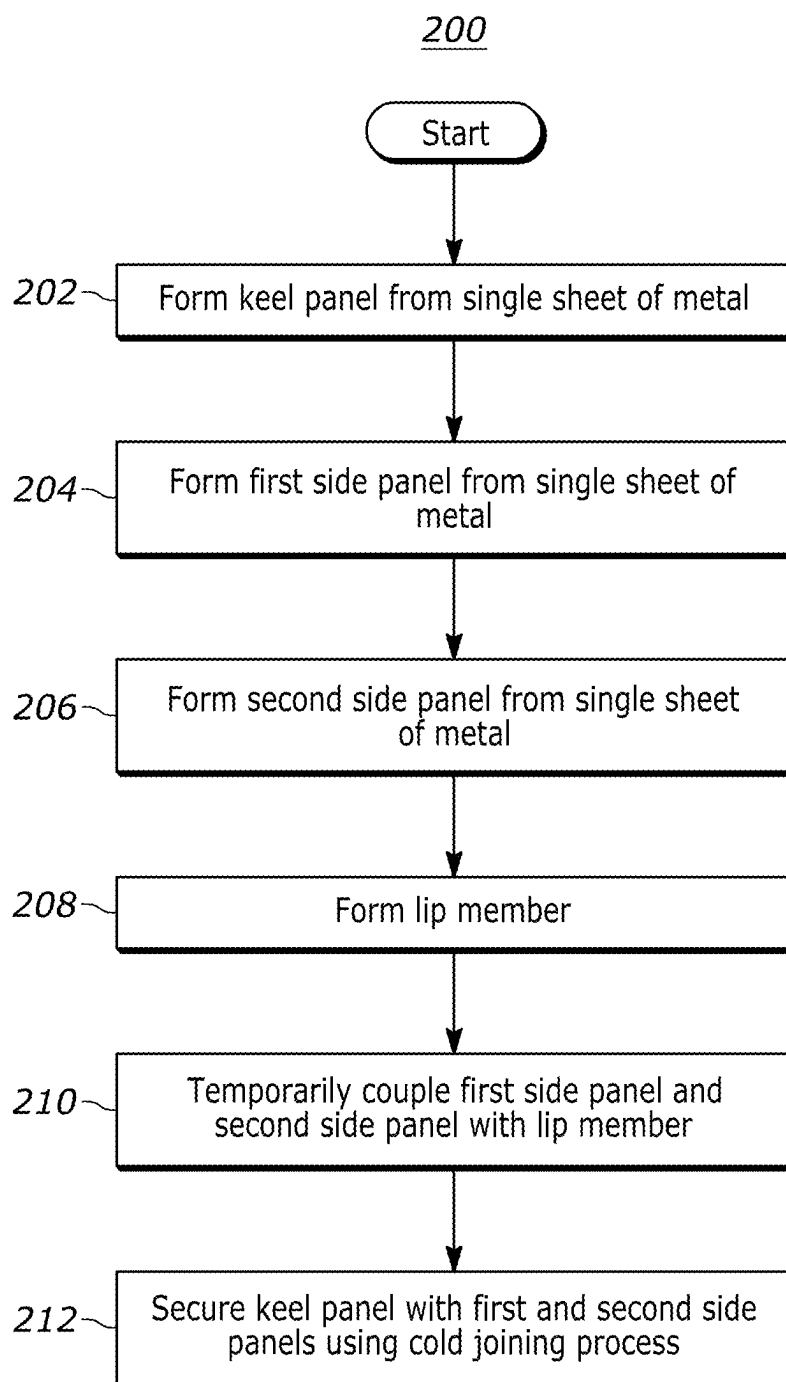


FIG. 11

SYSTEMS AND APPROACHES FOR ASSEMBLING A MARITIME VEHICLE

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application No. 63/561,282, filed on Mar. 4, 2024, the entirety of which is herein expressly incorporated by reference.

FIELD OF THE DISCLOSURE

[0002] This disclosure relates generally to maritime vehicles, and, more particularly, to approaches for assembling a maritime vehicle.

BACKGROUND

[0003] Maritime vehicles, or vehicles designed for use on or in the water, are commonly used for transportation, recreation, defense, scientific research, and other purposes. Examples of maritime vehicles include boats, watercraft, submarines, and amphibious vehicles. Maritime vehicles can be manned (i.e., operated by an onboard human) or unmanned, and unmanned maritime vehicles can be remotely controlled or can be autonomous.

[0004] Maritime vehicles may be constructed from a number of materials such as metals and/or polymers. When constructing a maritime vehicle using metals, steel may be used for its strength and ease of manufacturing. However, steel is a relatively heavy material, and as such, its use may be undesirable. Aluminum may also be used to produce maritime vehicles due its relatively lighter weight. However, because aluminum may be easier to damage relative to steel, assembling maritime vehicles using aluminum is oftentimes significantly more time consuming due to the need to account for its susceptibility to curl, bend, or otherwise deform. Such deformations may impact respective tolerances of structural components, thus leading to lengthy manufacturing times that are both labor and resource intensive.

[0005] Accordingly, there is a need for improved devices having improved functionalities.

SUMMARY

[0006] In accordance with a first aspect, an approach for assembling a maritime vehicle includes forming a keel panel from a single sheet of metal, forming a first side panel from a single sheet of metal, forming a second side panel from a single sheet of metal, and forming a lip member. The first side panel and the second side panel are temporarily coupled with the lip member. The keel panel is secured with the first side panel and the second side panel using a cold joining process.

[0007] In some examples, the method further includes fixturing the first and second side panels with the lip member via a riveting process. In some examples, the cold joining process may include an aluminum bonding process.

[0008] In some approaches, the method may further include the step of heat treating at least a portion of at least one of the keel panel, the first panel, or the second panel. In these and other approaches, the method may further include the step of coupling at least a portion of the keel panel with at least one of the first side panel or the second side panel via a spot welding process.

[0009] In some forms, the keel panel, the first panel, and the second panel may be formed from a single sheet of aluminum. In some forms, the keel panel, the first panel, and the second panel may be formed via a stamping process using a machine press.

[0010] In accordance with a second aspect, a method for assembling a maritime vehicle includes providing a keel panel, a first side panel, and a second side panel. Each of the keel panel, the first side panel, and the second side panel are constructed from a single sheet of metal. The first side panel and the second side panel are temporarily coupled with the lip member. The keel panel is secured with the first and second side panels using a cold joining process.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate examples of concepts that include the claimed invention, and explain various principles and advantages of those examples.

[0012] FIG. 1A is a top perspective view of an example of a maritime vehicle constructed in accordance with the teachings of the present disclosure;

[0013] FIG. 1B is a front view of the maritime vehicle of FIG. 1A;

[0014] FIG. 1C is a rear view of the maritime vehicle of FIG. 1A;

[0015] FIG. 1D is a bottom perspective view of the maritime vehicle of FIG. 1A;

[0016] FIG. 1E is another bottom perspective view of the maritime vehicle of FIG. 1A;

[0017] FIG. 1F is similar to FIG. 1A, but with the cap and various components of the maritime vehicle removed for illustrative purposes;

[0018] FIG. 1G is similar to FIG. 1F, but with additional components of the maritime vehicle removed for illustrative purposes;

[0019] FIG. 1H is a rear view of the maritime vehicle of FIG. 1F;

[0020] FIG. 1I is a first cross-sectional view taken along line I-I in FIG. 1A;

[0021] FIG. 1J is a second cross-sectional view taken along line J-J in FIG. 1A;

[0022] FIG. 1K is similar to FIG. 1A, but with the latching assembly of the maritime vehicle removed for illustrative purposes;

[0023] FIG. 2 is a top plan view of a portion of the example maritime vehicle of FIGS. 1A-1K;

[0024] FIG. 3 is an exploded perspective view of a portion of the example maritime vehicle FIGS. 1A-2;

[0025] FIG. 4 is a perspective view of example bulkheads for use in the example maritime vehicle of FIGS. 1A-3;

[0026] FIG. 5 is a perspective view of an example bulkhead and an example stringer support member for use in the example maritime vehicle of FIGS. 1A-4;

[0027] FIG. 6 is a close-up perspective view of an example forward bulkhead and aft tow point for use in the example maritime vehicle of FIGS. 1A-5;

[0028] FIG. 7 is a close-up perspective view of an example assembly process of an example bulkhead and stringer support for use in the example maritime vehicle of FIGS. 1A-6;

[0029] FIG. 8 is a close-up perspective and rear elevation view of an example bilge system outlet for use in the example maritime vehicle of FIGS. 1A-7;

[0030] FIG. 9 is a cross-sectional view of an example keel cooler scoop for use in the example maritime vehicle of FIGS. 1A-8;

[0031] FIG. 10 is a perspective view of the example maritime vehicle of FIGS. 1A-9 prior to sheet forming; and

[0032] FIG. 11 is a flowchart depicting an example method for assembling the maritime vehicle of FIGS. 1A-10.

[0033] Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions and/or relative positioning of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various examples. Also, common but well-understood elements that are useful or necessary in a commercially feasible examples are often not depicted in order to facilitate a less obstructed view of these various examples. It will further be appreciated that certain actions and/or steps may be described or depicted in a particular order of occurrence while those skilled in the art will understand that such specificity with respect to sequence is not actually required. It will also be understood that the terms and expressions used herein have the ordinary technical meaning as is accorded to such terms and expressions by persons skilled in the technical field as set forth above except where different specific meanings have otherwise been set forth herein.

[0034] The apparatus and method components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding examples of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

[0035] Although the figures show parts with clean lines and boundaries, some or all of these lines and/or boundaries may be idealized. In reality, the boundaries and/or lines may be unobservable, blended, and/or irregular. Use of terms such as up, down, top, bottom, side, end, front, back, etc. herein are used with reference to a currently considered or illustrated orientation. If they are considered with respect to another orientation, it should be understood that such terms must be correspondingly modified.

DETAILED DESCRIPTION

[0036] The present disclosure is directed to a maritime vehicle that is primarily intended for use for military purposes (e.g., for naval defense, patrolling waters and enforcing laws, reconnaissance, naval exploration, monitoring) but can also be used for other purposes if desired. The maritime vehicle is durable and configured to quickly, efficiently, and stealthily traverse a body of water once dispatched (e.g., from other maritime vehicles, beachheads, or an airdrop). The maritime vehicle is modular, with components that can be flexibly altered, removed, or added as desired in accordance with the mission of the maritime vehicle. The maritime vehicle can collaborate with other similar maritime vehicles and/or military assets when necessary. The maritime vehicle is preferably unmanned and autonomous though need not be.

[0037] FIGS. 1A-1K illustrate one example of a maritime vehicle 100 constructed in accordance with the teachings of the present disclosure. The maritime vehicle 100 is an unmanned vehicle configured to autonomously traverse a body of water. The maritime vehicle 100 generally includes a hull 104 and a cap 108 that is coupled to the hull 104 to secure various components within the maritime vehicle 100. The hull 104 is at least partially disposed in the body of water in which the maritime vehicle 100 is traversing. As will be discussed in detail below, the hull 104 is a mono-hull that has a front (or bow), a rear (or stern), a number of (e.g., two) side panels (i.e., a first side panel 120 and a second side panel 124), and a keel panel 130 coupled to another. The hull 104 is configured such that the hull provides a continuous planning surface that allows the maritime vehicle 100 to be highly maneuverable and to ride along the top of a body of water at high speeds, even in extreme weather conditions and difficult to navigate bodies of water. Meanwhile, the cap 108 is coupled to the hull 104 to cover and/or conceal the components of the maritime vehicle 100 disposed in and carried by the hull 104 as the maritime vehicle 100 traverses the body of water.

[0038] In this example, the hull 104 and the cap 108 each have a length that is equal to approximately 6 feet. In other examples, however, the length can vary. For example, the length can be equal to approximately 14 feet. The hull 104 is preferably entirely made of aluminum but can be partially or entirely be made of fiberglass and/or one or more other materials. In other examples, the maritime vehicle 100 can include two or more hulls (e.g., two parallel hulls). In this example, the cap 108 entirely covers the hull 104 (and the components therein). In other examples, however, the maritime vehicle 100 need not include the cap 108 or the cap 108 may only partially cover the hull 104 (and the components disposed therein).

[0039] In some examples, the cap 108 can be removably coupled to the hull 104 via a locking system. For example, as illustrated in FIGS. 1A-1K, the cap 108 can be removably coupled to the hull 104 via a plurality of latch mechanisms disposed around a perimeter of the maritime vehicle 100. Thus, the cap 108 can be removed to allow access to the interior of the hull 104. In other examples, however, the cap 108 can be permanently coupled to the hull 104.

[0040] The maritime vehicle 100 also includes a plurality of bulkheads 112 arranged within the hull 104. The bulkheads 112 divide the maritime vehicle 100 into a plurality of different compartments for receiving and retaining different components in the maritime vehicle 100.

[0041] The maritime vehicle 100 also includes a sensor system that is generally configured to collect data about various components of the maritime vehicle 100 as well as data about the environment surrounding the maritime vehicle 100 (including data about objects in that environment). To this end, the sensor system generally includes a plurality of sensors disposed on an exterior or an interior of the maritime vehicle 100. The sensors can include, for example, one or more pressure sensors (e.g., positioned to detect the pressure of the ambient air external to the maritime vehicle 100, the pressure of the water in which the maritime vehicle 100 is disposed, the pressure within the maritime vehicle 100), one or more temperature sensors (e.g., positioned to measure a temperature of a component of the maritime vehicle 100, a temperature of ambient air external to the maritime vehicle 100, a temperature of water

in which the maritime vehicle **100** is disposed), one or more acoustic sensors (e.g., sonar sensors), one or more LIDAR sensors, one or more location sensors (e.g., GPS sensors, compass sensors), one or more motion sensors (e.g., accelerometers, gyroscopes), one or more infrared sensors, one or more water level sensors, one or more humidity sensors, one or more power sensors (e.g., configured to detect charging or fueling levels), one or more lighting sensors (e.g., daylight sensors), one or more imaging sensors (e.g., CCD sensors, CMOS sensors), one or more magnetic sensors, or combinations thereof.

[0042] The maritime vehicle **100** also includes a power system **120** that is generally configured to power the maritime vehicle **100** (and the components of the maritime vehicle **100**). The power system **120** generally includes a propulsion system and one or more power sources **122** configured to power the propulsion system (and the other components within the maritime vehicle **100**). The propulsion system is generally configured to propel the maritime vehicle **100** in/on the water. The one or more power sources **122** can include, for example, one or more batteries, fuel (e.g., gasoline, diesel) stored in tanks carried by the maritime vehicle **100**, hydrogen stored in hydrogen tanks carried by the maritime vehicle **100**, solar panels (e.g., mounted to an exterior of the vehicle **100**), or other sources. The maritime vehicle **100** illustrated in FIGS. 1A-1K includes four battery assemblies each including a rechargeable battery. The maritime vehicle **100** generally also includes a cooling system configured to cool the propulsion system and/or the one or more energy sources, thereby preventing these components from overheating and leading to failure of the maritime vehicle **100**.

[0043] In operation, the maritime vehicle **100** may be used to deploy and/or retrieve payloads such as, for example, persons, weapons (e.g., drones, missiles, mines, bombs), cargo (e.g., food), scientific instruments, or other equipment. Payloads can be deployed aerially (into the air), underwater, or on the surface of the water. Payloads can also be retrieved from the air, from underwater, or the surface of the water. Payloads to be deployed can be disposed in the hull **104**, attached to the exterior surface of the hull **104**, or attached to the exterior surface of the cap **108** prior to deployment. Likewise, retrieved payloads can be stored in the hull **104**, attached to and stored on the exterior surface of the hull **104**, or attached to and stored on the exterior surface of the cap **108**.

[0044] The maritime vehicle **100** can also include other systems to help with the operation of the maritime vehicle **100**, namely a ballast system, a navigation system, and a vision system. The ballast system is generally configured to stabilize the maritime vehicle **100** in the water, regardless of whether the maritime vehicle **100** is stationary or on the move. To this end, the maritime vehicle **100** may include one or more ballast tanks or chambers selectively filled with water or air to vary the buoyancy of the maritime vehicle **100**. Alternatively, or additionally, the ballast system may include and utilize one or more inflatable devices to vary the buoyancy of the maritime vehicle **100**. The ballast system may also provide for the selective submerging and resurfacing of the maritime vehicle **100** in a similar manner. The navigation system, which may for example be an inertial navigation system, utilizes the sensors of the sensor system to track the position and orientation of the maritime vehicle **100** and to guide the maritime vehicle **100** to its

desired location in the body of water (or in a different body of water). The vision system is generally configured to capture, process, and analyze images obtained by the one or more image sensors and other data (e.g., data obtained by other sensors in the sensor system). The vision system can in turn identify or classify the environment surrounding the maritime vehicle **100** (including objects in that environment). Finally, the payload control system is configured to deploy or retrieve payloads.

[0045] The maritime vehicle **100** further includes a communications system that is generally configured to facilitate communication (a) between the maritime vehicle **100** and one or more central (remote) controllers, (ii) between the maritime vehicle **100** and and/or one or more other maritime vehicles **100** and/or other military assets (e.g., planes, ships), and (iii) between different components of the maritime vehicle **100**. The communications system generally includes one or more local controllers and one or more communication modules (e.g., one or more antennae, one or more receivers, one or more transmitters, one or more radios, one or more ethernet switches) to effectuate wired or wireless communication between the maritime vehicle **100** and the central controller(s) or other maritime vehicles **100**. For example, the maritime vehicle **100** includes a plurality of antennae disposed on an exterior of the cap **108** as well as a plurality of antennae disposed in the hull **104**.

[0046] The one or more local controllers are generally configured to communicate data (e.g., operational instructions, data from the sensor system, data from other maritime vehicles **100** or military assets) and to perform automated operations of the maritime vehicle **100** based on that data. In some examples, the maritime vehicle **100** includes a plurality of different local controllers. For example, the maritime vehicle **100** can include one or more propulsion controllers (for controlling the operation of the propulsion system), one or more sensor controllers (for controlling the sensors in the sensor system), one or more payload controllers (for deploying or retrieving payloads), one or more navigation controllers (as part of the navigation system), and one or more ballast controllers (for controlling the ballast system). It will be appreciated that each of the one or more controllers may be implemented as hardware (e.g., processor, die, integrated device), software (e.g., non-transitory processor readable medium), and/or combinations thereof, in one or more devices (e.g., processor, chip, computer, tablet, mobile device).

[0047] While not explicitly described or illustrated herein, it will be appreciated that the maritime vehicle **100** includes several additional components. For example, the maritime vehicle **100** includes various sealing elements configured to provide seals between different components of the vehicle **100** (or between the vehicle **100** and the environment surrounding the vehicle **100**). As another example, the maritime vehicle **100** also includes various fasteners that help to couple the components of the maritime vehicle **100** together. As yet another example, the maritime vehicle **100** includes cabling that helps to communicatively couple components of the maritime vehicle **100** together. As yet another example, the maritime vehicle **100** includes various electrical components that help to operate the maritime vehicle **100**, e.g., one or more relay boards, one or more DC-DC converters.

[0048] With reference to FIGS. 2-11, an approach for assembling the maritime vehicle **100** is now described.

Generally speaking, all or portions of the hull **104** may be assembled via a cold joining process that does not rely on traditional welding approaches that have a propensity to deform or otherwise damage the joined components. Instead, by incorporating a cold joining process, the hull **104** may be constructed from fewer elements and in a more efficient manner.

[0049] In some examples, the hull **104** includes the previously noted keel panel **130**, the first side panel **120**, and the second side panel **124**. The hull **104** may additionally include a lip member **136**. The first and the second side panel **120**, **124**, the keel panel **130**, and the lip member **136** may be constructed from a thin sheet of aluminum or similar metal. During manufacturing, the first and the second panels **120**, **124** and the keel panel **130** may be formed via an aluminum press stamping process, which may create desired features such as cuts, holes, openings, and the like on each respective panel **120**, **124**, **130**. The lip member **136** may similarly be machined via any suitable process. Further, it is to be appreciated that in some examples, the hull **104** may be constructed primarily from two side panels in addition to a “central” panel that spans a portion of the bow area. Other examples are possible.

[0050] In some arrangements, the hull **104** may be assembled by first temporarily coupling the first and the second side panels **120**, **124** (which may be mirror images of each other) with the lip member **136** via any number of suitable approaches. More specifically, in some approaches, the lip member **136** may include a number of holes or punches to receive a number of rivets. Similarly, the first and the second side panels **120**, **124** may also include a corresponding number of holes or punches to receive those same rivets. The first and the second side panels **120**, **124** may be aligned with the lip member **136**, and the rivets may be temporarily coupled with these elements before being fastened. In such an arrangement, the first and second panels **120**, **124** may be slightly movable relative to each other and the lip member **136** to assist with proper alignment.

[0051] The keel panel **130** may then be coupled or otherwise affixed with the first and the second side panels **120**, **124**. In some examples, a jig machine may be used in conjunction with specific fixtures to hold the components in place during the coupling process. Such fixtures may match dedicating locating features designed into the hull panels that are sealed (e.g., spot welded) after assembly is complete. In some examples, a cold joining (e.g., chemical welding, aluminum bonding, etc.) process is used that does not increase the temperature of the joined material to a degree that would generate distortion. In a next step, portions of the keel panel **130** may be secured to the first and second side panels **120**, **124** via rivets in addition to the cold joining process.

[0052] The first and the second side panels **120**, **124** may then be secured with the lip member **136** by riveting the components together. In some arrangements, a “true test” may be performed in which the hull **104** is turned upside-down such that the lip member **136** is placed on a generally flat surface to ensure all or substantially all of the lip member **136** is flush with the generally flat surface. This test may be performed prior to the step of riveting the first and the second side panels **120**, **124** to the lip member **136**.

[0053] It is to be appreciated that the hull **104** may include critical regions. In such areas, portions of the hull **104** may be heat treated or toleranced using any number of suitable

approaches and may subsequently be adhered via a spot welding process. While such a spot welding process may be relatively time-consuming compared with cold joining approaches, a limited number of areas may warrant this assembly process, and as a result, may not significantly increase overall assembly times. As a non-limiting example, the curved area of the hull **104** may be heat treated. Other examples are possible.

[0054] Once the chemical bonds of the hull **104** have properly cured, the remainder of the maritime vehicle **100** may be assembled. More specifically, the bulkheads **112**, support members **106**, and other supporting elements may be secured with the hull **104**, and the desired systems and subsystems may then be installed in the hull **104**. These bulkheads **112** and other supporting elements may be located via tab and slot through features (not illustrated) and may be sealed after assembly via any number of suitable approaches). In some examples, the hull **104** may include any number of tabs that may provide fixturing points during assembly, but may subsequently be removed prior to or after coupling the lip member **136** therewith.

[0055] Notably, such an assembly process may use the dimensions and (reduced) tolerances of the lip member **136** as a primary reference point for positioning of structural and other elements on the maritime vehicle **100** during assembly. More specifically, because the cold joining process described herein reduces or otherwise eliminates deformation and/or damage to panels of the hull **104**, the hull **104** may have tighter dimensional tolerances.

[0056] With reference to FIG. **11**, an approach **200** for assembling a maritime vehicle is provided. At a step **202**, the keel panel **130** is formed from a single sheet of metal. At a step **204**, the first side panel **120** is formed from a single sheet of metal. At a step **206**, the second side panel **124** is formed from a single sheet of metal. At a step **208**, the lip member **136** is machined (or, in some examples, cast, molded, extruded, formed, etc.). At a step **210**, the first and second side panels **120**, **124** are temporarily coupled with the lip member **136**. At a step **212**, the keel panel **130** is secured with the first and second side panels **120**, **124** using a cold joining process.

[0057] It is to be appreciated that in some examples, a combination of cold joining and hot joining processes may be used to assemble the maritime vehicle **100**. As an example, the lip member **136** may be attached to the first and second side panels **120**, **124** via a cold joining process, while other steps of the maritime vehicle **100** may be assembled using hot forming processes. The combination of these processes (e.g., adhesive or cold joining and mechanical reinforcements in the form of rivets) may advantageously provide for improved shock performance and longevity.

[0058] So arranged, the approaches for assembling a maritime vehicle **100** described herein may significantly reduce overall assembly times. As an example, the approaches described herein may result in assembly times between approximately seven and 10 hours, whereas previous approaches may require assembly times between approximately 35-40 hours. In the current approaches, much of the assembly time is driven by curing times, whereas in previous approaches, the assembly time was dictated by the time needed to form and bend various panels while avoiding deformation and non-uniform geometries.

[0059] Finally, although certain maritime vehicles have been described herein in accordance with the teachings of

the present disclosure, the scope of coverage of this patent is not limited thereto. On the contrary, while the invention has been shown and described in connection with various preferred embodiments, it is apparent that certain changes and modifications, in addition to those mentioned above, may be made. This patent covers all embodiments of the teachings of the disclosure that fairly fall within the scope of permissible equivalents. Accordingly, it is the intention to protect all variations and modifications that may occur to one of ordinary skill in the art.

[0060] In the foregoing specification, specific embodiments have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the invention as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present teachings. Numerous alternative examples could be implemented, using either current technology or technology developed after the filing date of this patent, which would still fall within the scope of the claims. Additionally, the described embodiments/examples/implementations should not be interpreted as mutually exclusive and should instead be understood as potentially combinable if such combinations are permissive in any way. In other words, any feature disclosed in any of the aforementioned embodiments/examples/implementations may be included in any of the other aforementioned embodiments/examples/implementations.

[0061] The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims. The claimed invention is defined solely by the appended claims including any amendments made during the pendency of this application and all equivalents of those claims as issued.

[0062] Moreover, in this document, relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” “has,” “having,” “includes,” “including,” “contains,” “containing” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises, has, includes, contains a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a,” “has . . . a,” “includes . . . a,” “contains . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises, has, includes, contains the element. The terms “a” and “an” are defined as one or more unless explicitly stated otherwise herein. The terms “substantially,” “essentially,” “approximately,” “about” or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting example the term is defined to be within 10%, in another example within 5%, in another example within 1% and in another example within 0.5%. The term “coupled” as used herein is defined as connected, although not necessarily directly and not necessarily

mechanically. A device or structure that is “configured” in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

[0063] Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, “A, B or C” refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, and (7) A with B and with C. As used herein, the phrase “at least one of A and B” is intended to refer to any combination or subset of A and B such as (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. Similarly, the phrase “at least one of A or B” is intended to refer to any combination or subset of A and B such as (1) at least one A, (2) at least one B, and (3) at least one A and at least one B.

[0064] The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various examples for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed examples require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter may lie in less than all features of a single disclosed example. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

[0065] Additionally, the benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims.

[0066] Finally, any references, including, but not limited to, publications, patent applications, and patents cited herein are hereby incorporated in their entirety by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

[0067] The patent claims at the end of this patent application are not intended to be construed under 35 U.S.C. § 112(f) unless traditional means-plus-function language is expressly recited, such as “means for” or “step for” language being explicitly recited in the claim(s). The systems and methods described herein are directed to an improvement to computer functionality, and improve the functioning of conventional computers.

[0068] Although certain example methods, apparatus and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all methods, apparatus and articles of manufacture fairly falling within the scope of the claims of this patent.

What is claimed is:

1. A method for assembling a maritime vehicle, the method comprising:

- forming a keel panel from a single sheet of metal;
- forming a first side panel from a single sheet of metal;
- forming a second side panel from a single sheet of metal;
- forming a lip member;
- temporarily coupling the first side panel and the second side panel with the lip member;

securing the keel panel with the first side panel and the second side panel using a cold joining process.

2. The method of claim 1, further including fixturing the first side panel and the second side panel with the lip member via a riveting process.

3. The method of claim 1, wherein the step of securing the keel panel with the first and second side panels using a cold joining process comprises using an aluminum bonding process.

4. The method of claim 1, further comprising the step of heat treating at least a portion of at least one of the keel panel, the first side panel, or the second side panel.

5. The method of claim 4, further comprising the step of coupling at least a portion of the keel panel with at least one of the first side panel or the second side panel via a spot welding process.

6. The method of claim 1, wherein the keel panel, the first side panel, and the second side panel are formed from a single sheet of aluminum.

7. The method of claim 1, wherein the keel panel, the first side panel, and the second side panel are each formed via a stamping process using a machine press.

8. A method for assembling a maritime vehicle, the method comprising:

providing a keel panel, a first side panel, and a second side panel, each of the keel panel, the first side panel, and the second side panel being constructed from a single sheet of metal;

temporarily coupling the first side panel and the second side panel with a lip member;

securing the keel panel with the first side panel and the second side panel using a cold joining process.

9. The method of claim 8, further including fixturing the first side panel and the second side panel with the lip member via a riveting process.

10. The method of claim 8, wherein the step of securing the keel panel with the first and second side panels using a cold joining process comprises using an aluminum bonding process.

11. The method of claim 8, further comprising the step of heat treating at least a portion of at least one of the keel panel, the first side panel, or the second side panel.

12. The method of claim 11, further comprising the step of coupling at least a portion of the keel panel with at least one of the first side panel or the second side panel via a spot welding process.

13. The method of claim 8, wherein the keel panel, the first side panel, and the second side panel are formed from a single sheet of aluminum.

14. The method of claim 8, wherein the keel panel, the first side panel, and the second side panel are each formed via a stamping process using a machine press.

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